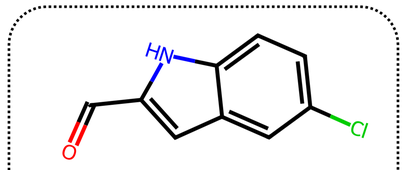
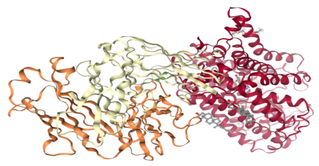
The overview of HAG-DTA



Drug SMILES Input



Protein sequence Input

Hierarchical Network

H_1 Block → H_2 Block

Deep Graph Network

CNN Block

Self-Attention

Self-Attention

Self-Attention

x_{H_1} x_{H_2} MMD loss x_G x_P

Attention Block

MLP

Y

x_{H_1} : Functional group features

x_{H_2} : Molecular Fragment features

x_G : Global features

x_P : Protein features

Deep Graph Network

Input

GNN

Graph Norm

GNN

Graph Norm

GNN

Graph Norm

GNN

Graph Norm

GNN

Graph Norm

Output

CNN Block

Input

Conv1d 5×128 → Max Pool

Conv1d 4×128 → Max Pool

Conv1d 3×128 → Max Pool

MLP Block

Output

Attention Block

Input: x_G x_{H_1} x_{H_2} x_P

MLP Block → a_G

MLP Block → a_{H_1}

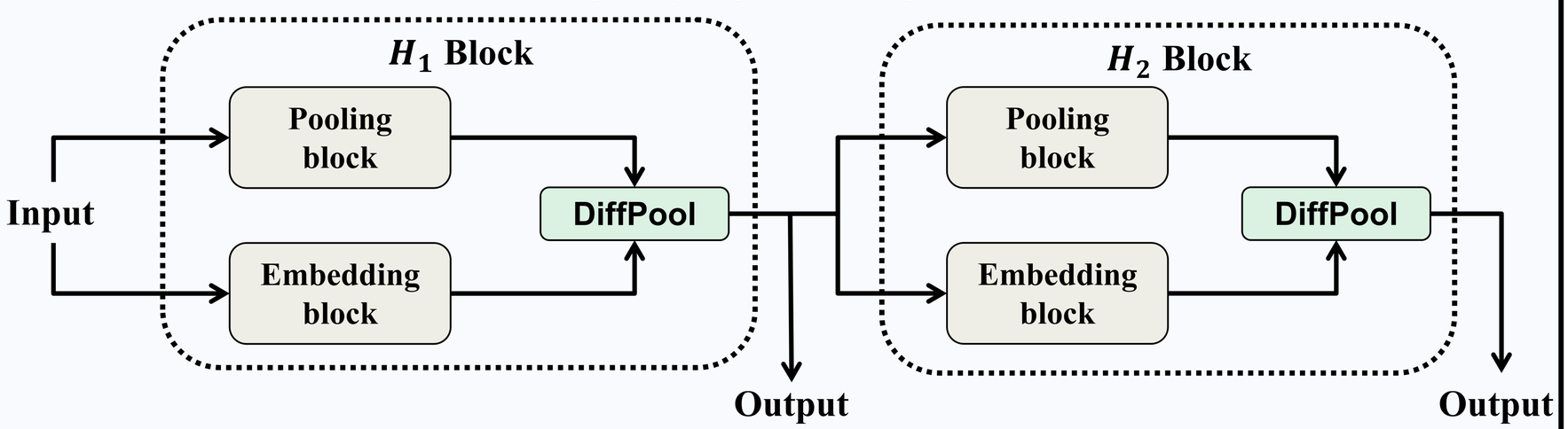
MLP Block → a_{H_2}

MLP Block → a_P

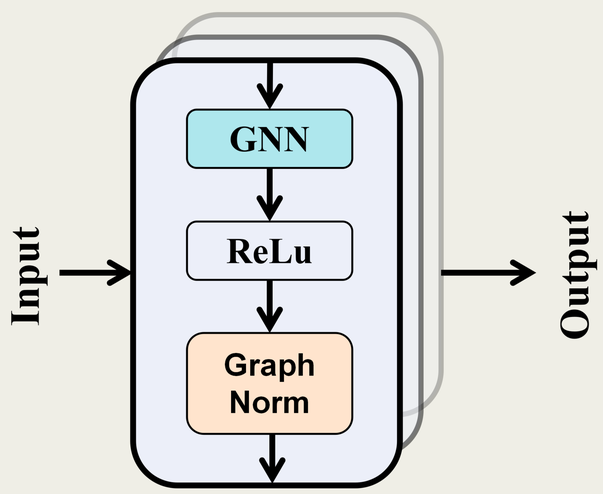
Input → ⊕

Output

Hierarchical Network



H_1 Block : First-level Hierarchical network

H_2 Block : Second-level Hierarchical network

Pooling/Embedding block